INSTITUTE AND FACULTY OF ACTUARIES

EXAMINATION

11 September 2024 (am)

Subject CS1 – Actuarial Statistics
Core Principles

Paper A

Time allowed: Three hours and twenty minutes

In addition to this paper you should have available the 2002 edition of the Formulae and Tables and your own electronic calculator.

If you encounter any issues during the examination please contact the Assessment Team on T. 0044 (0) 1865 268 873.

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1 An insurance company sells policies that provide coverage for 1 year for car accidents. The number of accidents per policy follows a Poisson distribution with a mean of 0.5 accidents per policy. Let X be the number of accidents in a randomly selected policy.

(i) Calculate the probability that X is at least 2. [3]

The same insurance company proposes a life insurance policy that pays out a benefit on the death of the policyholder. The lifetime of the policyholder is assumed to follow an exponential distribution with a mean of 75 years.

(ii) Calculate the conditional probability that the policyholder will live beyond the age of 80, given that they have already reached the age of 70. [4]

[Total 7]

2 In a Bayesian analysis a sample $x_1, \ldots, x_n$ is drawn from an exponential distribution with unknown parameter λ. Prior knowledge about λ is available and a prior distribution is chosen with probability density function $p(\lambda) = k\lambda^{k-1}$ for $0 \leq \lambda \leq 1$ and $p(\lambda) = 0$ for $\lambda < 0$ or $\lambda > 1$. The parameter k is a positive integer.

(i) Identify which **one** of the following expressions gives the Bayesian estimate for λ under all-or-nothing loss:

A $\hat{\lambda} = \min\left(1, \frac{n+k-1}{\sum_{i=1}^{n} x_i}\right)$

B $\hat{\lambda} = \min\left(1, \frac{n+k}{\sum_{i=1}^{n} x_i}\right)$

C $\hat{\lambda} = \min\left(1, \frac{\sum_{i=1}^{n} x_i}{n+k-1}\right)$

D $\hat{\lambda} = \min\left(1, \frac{\sum_{i=1}^{n} x_i}{n+k}\right)$.

[3]

(ii) Comment on why the answer in part (i) needs to involve the minimum. [1]

[Total 4]

3 An actuary working at an investment bank is responsible for a portfolio of 300 corporate bonds. The number of defaults in the portfolio in a given year is assumed to follow a binomial distribution with parameter p. In this portfolio, five bonds defaulted in the previous year.

It is assumed that the prior distribution for the probability of default of a single bond, p, is a beta distribution. External analysis suggests that p has a mean of 0.6 and a standard deviation of 0.2.

(i) Verify that the parameters of the beta prior distribution are: $\alpha = 3$ and $\beta = 2$. [3]

The actuary determines that the credibility factor Z can be derived using the following equation:

$$Z = \frac{n}{n + \alpha + \beta}$$

where n is the number of bonds in the portfolio.

(ii) Calculate the credibility factor Z. [1]

(iii) Calculate the Bayesian credibility estimate for the probability of default. [2]

(iv) Comment on your answers to parts (ii) and (iii). [2]

[Total 8]

4 A wooden stick is broken at a particular point, chosen at random, along its length. The length X, of the shorter piece follows a uniform distribution. You may assume that $X \leq \frac{L}{2}$, where L is the length of the wooden stick.

(i) State the probability $P(X < x)$ in terms of x and L. [1]

Let R be the ratio of the length of the shorter piece to the length of the longer piece.

(ii) Show that the probability that the ratio, R, is less than r, is:

$$P(R < r) = \frac{2r}{1+r}$$

where r is a given positive number. [3]

(iii) Justify that the probability in part (ii) satisfies the conditions for a cumulative distribution function. [3]

(iv) Calculate the mean of R. [4]

(v) Identify which **one** of the following values gives the variance of R:

A 0.4971
B 0.0185
C 0.0782
D 0.1369.

[3]
[Total 14]

5 Let $X_1, X_2 \ldots X_n$ be a random sample from a distribution with probability density function given as:

$$f(x) = \theta(1+x)^{-(\theta+1)}, \text{ for } 0 < x < \infty$$

where $0 < \theta < \infty$ is an unknown parameter.

Consider a gamma prior distribution for θ with parameters $a, b > 0$,
i.e. $\theta \sim$ Gamma(a,b).

(i) Demonstrate that the posterior distribution of θ is Gamma$\left(a+n, b+\sum_{i=1}^{n} \ln(1+x_i)\right)$. [6]

(ii) Comment on the relationship between the prior and posterior distribution of θ. [1]

A random sample gives $\sum_{i=1}^{15} \ln(1+x_i) = 5.1$ and the prior distribution for θ is assumed to be Gamma(1,0.4).

(iii) Determine the Bayes estimate of parameter θ under the quadratic loss function. [2]

(iv) Determine a 95% equal-tailed credible interval for θ. [4]

[Total 13]

6 A data science actuary is analysing the relationship between the age of a policyholder (x) and the number of claims that the policyholder made over the past year (y). The actuary collects data from 100 policyholders and performs a linear regression analysis.

The linear regression model is given by:

$$y = a + bx + e$$

(i) Explain what a, b and e represent in the regression model. [2]

The results of the regression analysis are given as

$$\hat{a} = 60, \hat{b} = -0.5, SS_{RES} = 2521.416 \text{ and } S_{yy} = 17970.44$$

(ii) Comment on the values of $\hat{a}$ and $\hat{b}$. [2]

(iii) Calculate the coefficient of determination for this regression model. [2]

(iv) Comment on the coefficient of determination calculated in part (iii). [1]

You are given $S_{xx} = 67844$ and $\bar{x} = 61.8$.

(v) Determine a 95% confidence interval for the coefficient b. [3]

(vi) Perform a statistical test for the hypotheses $b = 0$ versus $b \neq 0$ at 5% level using your answer to part (v). [1]

(vii) Determine a 95% prediction interval for the number of claims made over the past year for a 50 year old policyholder. [4]

[Total 15]

7 An art dealer is evaluating the performance of an art valuation expert. For this purpose, the dealer has collected a sample of expert valuations (in £1,000s), V_i, and achieved auction prices (in £1,000s), P_i, of ten paintings. The valuations and achieved auction prices are given in the following table together with the differences, $D_i = P_i - V_i$, and relative differences, $R_i = \frac{D_i}{P_i}$ (relative to the achieved price). The paintings are ordered according to the value suggested by the expert.

Valuation V_i	89	95	123	149	158	239	244	294	327	365
Price P_i	68.1	113.4	155.8	172.6	70.3	399.2	369.4	526.5	450.9	473.7
$D_i = P_i - V_i$	−20.9	18.4	32.8	23.6	−87.7	160.2	125.4	232.5	123.9	108.7
$R_i = D_i/P_i$	−0.307	0.162	0.211	0.137	−1.248	0.401	0.339	0.442	0.275	0.229

$$\sum_{i=1}^{10} D_i = 716.9, \sum_{i=1}^{10} R_i = 0.641, \sum_{i=1}^{10} D_i^2 = 132{,}712, \sum_{i=1}^{10} R_i^2 = 2.3404$$

(i) Calculate the mean and sample standard deviation of the differences D_i. [3]

(ii) Calculate a 95% confidence interval for the mean difference between achieved auction price and valuation. [2]

(iii) Perform a test of the null hypothesis that the mean difference between achieved auction price and valuation is zero against the alternative that the mean difference is not zero based on your answer to part (ii). Use a significance level of 5% for your test decision. [2]

(iv) Comment on the assumption required for the confidence interval in part (ii) and the test in part (iii) to be appropriate. [2]

It is suggested that the differences D_i are not a suitable quantity to judge if the expert is biased. Instead, relative differences should be used.

(v) Perform a test of the null hypothesis that the mean relative difference is zero against the alternative that the mean relative difference is not zero. Use a significance level of 5%. [4]

(vi) Comment on your answers in parts (iii) and (v). [2]

(vii) Determine a 95% prediction interval for the price that can be achieved at an auction for a painting that is valued at £200 by the expert. Base the prediction interval on the sample $R_1, \ldots, R_{10}$ of the relative difference. As an approximation you may assume that the $R_1, \ldots, R_{10}$ are independent of each other and normally distributed with unknown mean but known standard deviation of 0.25. [5]

[Total 20]

8 An actuary working for an insurance company wishes to model the size of claims on a particular type of insurance policy, based on the recorded sizes of past claims. The insurance policy has a limit, M, which is the maximum amount the insurance company will pay for a single claim. Whereas sizes of claims are unlimited, recorded sizes of claims are limited to M.

Let X_i be a random variable denoting the size of claim i, where all $\{X_i\}$ are independent. The actuary wishes to model the size of claims using an exponential distribution with parameter λ. The collected data consists of $n + k$ claims where n is the number of claims whose recorded size is below the limit and k is the number of claims whose recorded size equals the limit.

(i) Explain why the collected data is an example of censored data. [2]

(ii) Show that the log-likelihood function is given by:

$$l(\lambda; x) = n \cdot \ln(\lambda) - \lambda\left(\mathrm{kM} + \sum_{i=1}^{n} x_i\right)$$

[5]

(iii) Derive the maximum likelihood estimator, $\hat{\lambda}$ of λ. In your answer you should verify that it maximises the likelihood function. [4]

For the insurance policy that the actuary is considering, the limit is £50,000. Of the past 300 recorded claims, nine were at the limit and the **remaining** claims totalled £750,000.

(iv) Calculate the maximum likelihood estimate of λ for the given data. [2]

(v) Calculate, using your answer to part (iv), the maximum likelihood estimate of the mean claim size. [1]

The actuary wishes to obtain a confidence interval for the mean claim size.

(vi) Calculate the Cramer–Rao Lower Bound of $\hat{\lambda}$. [2]

(vii) Calculate, using your answer to part (vi), an approximate 95% confidence interval for the mean claim size. [3]

[Total 19]

END OF PAPER